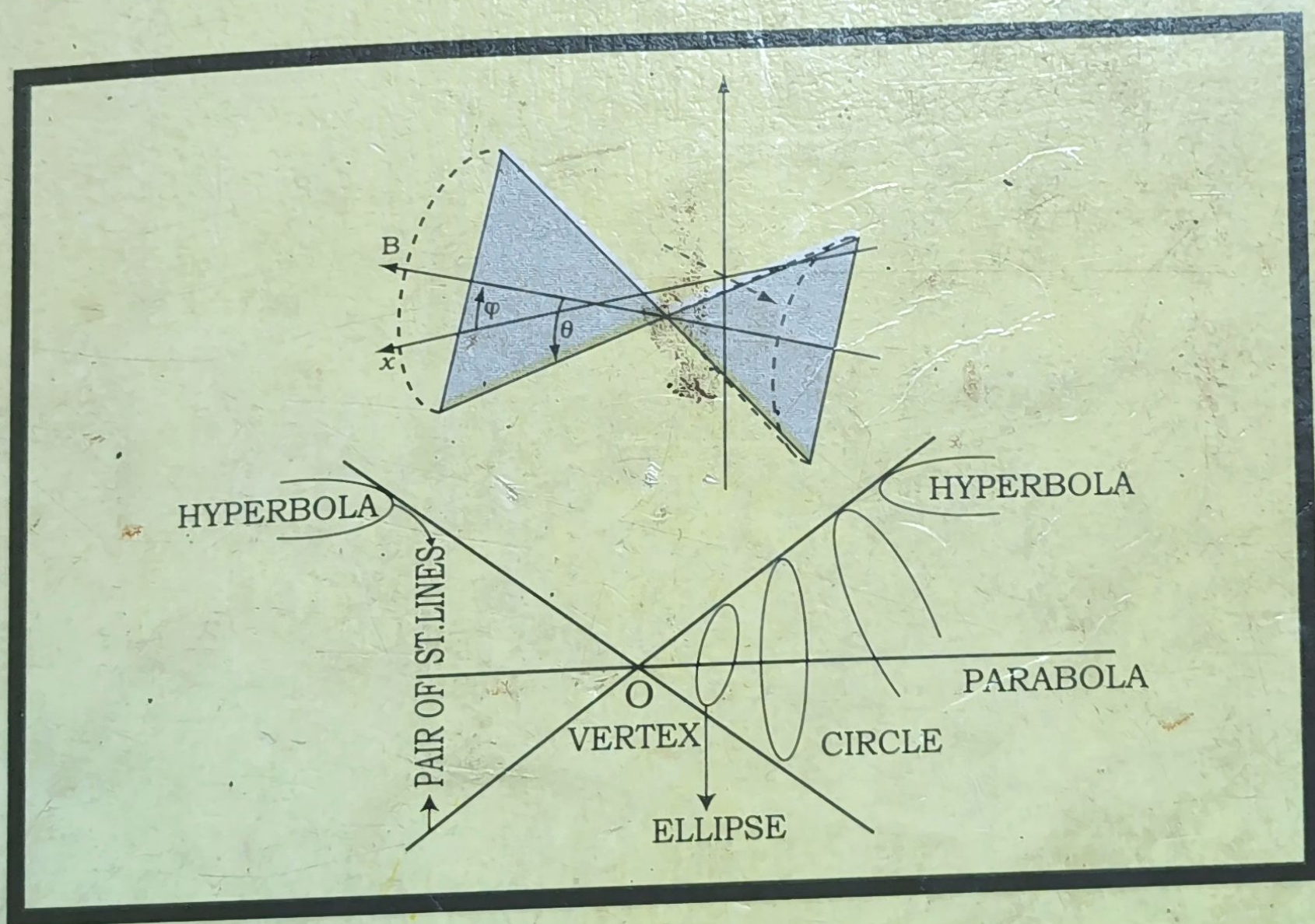


**A TEXT BOOK
ON
CO-ORDINATE GEOMETRY
(TWO AND THREE DIMENSIONS)
WITH VECTOR ANALYSIS**



RAHMAN & BHATTACHARJEE

Centroid of the triangle ABC is

$$G\left(\frac{a+0+0}{3}, \frac{0+b+0}{3}, \frac{0+0+c}{3}\right) \text{ i.e. } \left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$$

But we are given that the centroid is the point (p, q, r)

$$\therefore p = \frac{1}{3}a, q = \frac{1}{3}b, r = \frac{1}{3}c$$

$$a = 3p, b = 3q, c = 3r$$

$$\text{Therefore eq. (1) is } \frac{x}{3p} + \frac{y}{3q} + \frac{z}{3r} = 1 \text{ or } \frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 3$$

Ex. 11. Find the distance of the point $(2, 0, 1)$ and $(3, -3, 2)$ from the plane $x - 2y + z = 6$ and find whether the two points lie on the same side or opposite sides of the plane.

$$\text{The equation of the plane is } x - 2y + z - 6 = 0 \quad (1)$$

$$\text{Its distance from } (2, 0, 1) \text{ is } \frac{2 - 2.0 + 1 - 6}{\sqrt{1^2 + 2^2 + 1^2}} = \frac{-3}{\sqrt{6}}$$

$$\text{Its distance from } (3, -3, 2) \text{ is } \frac{3 - (2 \cdot -3) + 2 - 6}{\sqrt{1^2 + 2^2 + 1^2}} = \frac{5}{\sqrt{6}}$$

The distance $3/\sqrt{6}$ and $5/\sqrt{6}$

Put the point $(2, 0, 1)$ in the L.H.S of eq. (1) this gives -3 , i.e negative value.

Put point $(3, -3, 2)$ in this L.H.S in eq. (1), this gives 5 , i.e positive value.

The two results are of opposite signs. Therefore the two points lie on opposite sides of the plane.

Ex. 12. A variable plane is at a constant distance p from the origin and meets the axis in A, B, C. Through A, B, C, planes are drawn parallel to the co-ordinate planes. Show that the locus of their point of intersection is $x^2 + y^2 + z^2 = p^2$

Let the equation of the variable plane be $lx + my + nz = p$.

Here l, m, n vary but p is constant as the plane is always at constant distance from the origin, l, m, n are the actual direction cosines of the plane.

It cuts the axis of x at $A(p/l, 0, 0)$ axis of y at $B(0, p/m, 0)$ and axis of z at $C(0, 0, p/n)$

Now the planes through A, B, C are parallel to the co-ordinate planes, then the co-ordinates of point of intersection $(p/l, p/m, p/n)$,

$$\therefore x = p/l, y = p/m, z = p/n$$

$$\text{Therefore, } p^2/x^2 + p^2/y^2 + p^2/z^2 = l^2 + m^2 + n^2 = 1$$

$$\text{or, } x^2 + y^2 + z^2 = p^2$$

Ex. 13. A plane makes intercepts $OA = a$, $OB = b$ and $OC = c$, on the axes of co-ordinates. Find the area of the triangle ABC made by the plane.

A_x = area of the plane triangle formed by the three points which are the projections of the vertices on the yz plane.

Clearly the points B and C are in the yz plane and the projection of the point $A(a, 0, 0)$ is $(0, 0, 0)$

Putting the values of k in (4) we get

$$2x - y + \frac{3}{14}(3z - y) = 0$$

$$\Rightarrow 28x - 14y + 9z - 3y = 0$$

$$\Rightarrow 28x - 17y + 9z = 0$$

Which is the required plane.

20. A point p moves on a fixed plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. the plane through p perpendicular to op meets the axes in A, B, C respectively. If the planes are drawn through A, B, C parallel to the co-ordinate planes then show that the locus of the point inter section is

$$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{ax} + \frac{1}{by} + \frac{1}{cz}$$

Solution : The given plane

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad \dots \dots \dots (1)$$

Let the point $P(\alpha, \beta, \gamma)$ passes through (1).

$$\therefore \frac{\alpha}{a} + \frac{\beta}{b} + \frac{\gamma}{c} = 1 \quad \dots \dots \dots (2)$$

Let equation of the plane through $P(\alpha, \beta, \gamma)$ perpendicular to op is

$$\alpha(x - \alpha) + \beta(y - \beta) + \gamma(z - \gamma) = 0$$

$$\Rightarrow \frac{\alpha x}{\alpha^2 + \beta^2 + \gamma^2} + \frac{\beta y}{\alpha^2 + \beta^2 + \gamma^2} + \frac{\gamma z}{\alpha^2 + \beta^2 + \gamma^2} = 1 \quad \dots \dots \dots (3)$$

Since the plane (3) meets the axes in A, B, C .

$$\therefore \text{Co-ordinates of } A \left(\frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, 0, 0 \right)$$

$$\text{----- } B \left(0, \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta}, 0 \right)$$

$$\text{----- } C \left(0, 0, \frac{\alpha^2 + \beta^2 + \gamma^2}{\gamma} \right)$$

The planes through A, B, C Parallel to the Co-ordinate planes intersect at Q (say)

$$\therefore \text{Co-ordinate of } Q \left(\frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta}, \frac{\alpha^2 + \beta^2 + \gamma^2}{\gamma} \right)$$

i.e. at the point Q

$$x = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, y = \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta}, z = \frac{\alpha^2 + \beta^2 + \gamma^2}{\gamma}$$

$$\text{Now, L.H.S} = \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \left(\frac{\alpha}{\alpha^2 + \beta^2 + \gamma^2} \right)^2 + \left(\frac{\beta}{\alpha^2 + \beta^2 + \gamma^2} \right)^2 + \left(\frac{\gamma}{\alpha^2 + \beta^2 + \gamma^2} \right)^2$$

$$= \frac{\alpha^2 + \beta^2 + \gamma^2}{(\alpha^2 + \beta^2 + \gamma^2)^2}$$

$$\therefore \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{\alpha^2 + \beta^2 + \gamma^2}$$

$$\begin{aligned} \text{R.H.S} &= \frac{1}{ax} + \frac{1}{by} + \frac{1}{cz} = \frac{\alpha}{a(\alpha^2 + \beta^2 + \gamma^2)} + \frac{\beta}{b(\alpha^2 + \beta^2 + \gamma^2)} + \frac{\gamma}{c(\alpha^2 + \beta^2 + \gamma^2)} \\ &= \frac{\alpha/a + \beta/b + \gamma/c}{\alpha^2 + \beta^2 + \gamma^2} \end{aligned}$$

$$\therefore \frac{1}{ax} + \frac{1}{by} + \frac{1}{cz} = \frac{1}{\alpha^2 + \beta^2 + \gamma^2} \text{ [By --(2)]}$$

$\therefore \text{R.H.S} = \text{L.H.S}$ (Proved)

21. Find the equation of the plane through the points (1, 0, -1) and (2, 1, 3) and perpendicular to the plane $2x + y + z = 1$. [N.U.H -2005]

Solutions: The given plane

$$2x + y + z = 1 \quad \dots \quad (1)$$

Let equation of the plane through the point (1, 0, -1) is

$$a(x-1) + b(y-0) + c(z+1) = 0 \quad \dots \quad (2)$$

It passes through (2, 1, 3).

$$\therefore a(2-1) + b(1-0) + c(3+1) = 0$$

$$\Rightarrow a + b + 4c = 0 \quad \dots \quad (3)$$

Since the (1) and (2) are perpendicular.

$$\text{So, } 2a + b + c = 0 \quad \dots \quad (4)$$

From (3) and (4), we get by the rule of cross multiplication

$$\frac{a}{\frac{1}{3} - \frac{4}{7}} = \frac{b}{\frac{8}{3} - 1} = \frac{c}{1 - 2}$$

$$\frac{a}{3} = \frac{b}{7} = \frac{c}{-1} = k \text{ (say)}$$

$$\therefore a = -3k, b = 7k, c = -k$$

Putting the values of a, b, c in (2) we get

$$(2) \Rightarrow -3k(x-1) + 7k(y-0) - k(z+1) = 0$$

$$3x - 7y + z - 2 = 0 \text{ Which is required plane.}$$

22. A variable plane passes through a fixed point (a, b, c) and meets the axes in A, B, C. Show that the locus of the point of intersection of the planes through A, B, C parallel to the Co-

ordinate planes is $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$

[N.U.H-2006]

Solution: Let the variable plane be

$$\frac{x}{a_1} + \frac{y}{b_1} + \frac{z}{c_1} = 1 \quad \dots \quad (1)$$

it passes through the point (a, b, c)

$$\therefore \frac{a}{a_1} + \frac{b}{b_1} + \frac{c}{c_1} = 1 \quad \dots \quad (2)$$

Again, plane (1) meets the axes in A, B, C.

∴ Co-ordinates of A ($a_1, 0, 0$)

----- B ($0, b_1, 0$)

----- C ($0, 0, c_1$)

Planes through A, B, C parallel to the Co-ordinate planes intersect at M (say)

∴ Co-ordinate of M (a_1, b_1, c_1)

At the point M; $a_1 = x, b_1 = y, c_1 = z$.

Putting the values of a_1, b_1, c_1 in (2) we get $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$ which is the required locus.

23. Two system of rectangular axes have the same origin. If a plane cuts the axes at a, b, c and a_1, b_1, c_1 respectively from the origin. Prove that $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a_1^2} + \frac{1}{b_1^2} + \frac{1}{c_1^2}$

Sol : There are two rectangular system,

Eq. of the plane w. r. to the system (1) and (2) and are

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad \dots \quad (1)$$

$$\text{and } \frac{x}{a_1} + \frac{y}{b_1} + \frac{z}{c_1} = 1 \quad \dots \quad (2)$$

Perpendicular distance from the origin to planes (1) and (2) are

$$d_1 = \frac{0 + 0 + 0 - 1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}} = \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

$$d_2 = \frac{0 + 0 + 0 - 1}{\sqrt{\frac{1}{a_1^2} + \frac{1}{b_1^2} + \frac{1}{c_1^2}}} = \frac{1}{\sqrt{\frac{1}{a_1^2} + \frac{1}{b_1^2} + \frac{1}{c_1^2}}}$$

Since Two system have same origin

$$\begin{aligned} \text{So } d_1 = d_2 &\Rightarrow \frac{0 + 0 + 0 - 1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}} = \frac{1}{\sqrt{\frac{1}{a_1^2} + \frac{1}{b_1^2} + \frac{1}{c_1^2}}} \\ &= \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a_1^2} + \frac{1}{b_1^2} + \frac{1}{c_1^2} \quad (\text{Proved}) \end{aligned}$$

24. Find the equation of the plane passing through the points (1, 2, -2) and (3, -2, 6) and perpendicular to the plane $2x - y - z + 7 = 0$ [N.U. H-2009]

Solution : The Given plane

$$2x - y - z + 7 = 0 \quad \dots \quad (1)$$

Let equation of the plane through the point (1, 2, -2) is

$$a(x - 1) + b(y - 2) + c(z + 2) = 0 \quad \dots \quad (2)$$

It passes through (3, -2, 6)